Physics of Complex Systems Homework 4

Jun Wei Tan*

Julius-Maximilians-Universität Würzburg

(Dated: June 8, 2026)

Problem 1 (Fireflies)**.** Imagine a huge cloud of fireflies (Glühwürmchen), each of them being at a fixed position $\vec{x}$ and randomly emitting light with probability $p(\vec{x})$.

You have made a snapshot with your camera, analyzing N of these fireflies in detail. Indexing them by $i = 1, \ldots, N$, you have produced a training data set $S = \{(\vec{x}_i, t_i)\}$, where $t_i = 1$ means that the firefly emits light and $t_i = 0$ means that it does not. Hence $p_i = \langle t_i \rangle$ is the probability of the firefly i to emit light. Now you have trained a feed-forward network to predict this probability as a function of $\vec{x}$.

(a) Which activation function do you have to use in the last layer of the network? Explain your answer.

(b) Let $\{q_i\}$ be the probability distribution predicted by the network. In machine learning, the cross entropy of the data set S is defined as

$$H_S = -\frac{1}{N} \sum_{i=1}^{N} \left(t_i \log_2(q_i) + (1 - t_i) \log_2(1 - q_i) \right),$$

where $\log_2$ denotes the logarithm to the base 2 and where the sum runs over all training examples. What would be the expectation value of $H = \langle H_S \rangle$ over infinitely many snapshots?

(c) Give a reasonable argument why, for a sufficiently large training data set, it is reasonable to assume that

$$H_S \approx H.$$

(d) Prove that H is minimal if the predicted probability distribution $\{q_i\}$ coincides with the true distribution $\{p_i\}$.

* jun-wei.tan@stud-mail.uni-wuerzburg.de

(e) Let H_{min} be the minimal value of H. Show that the difference

$$H - H_{\text{min}}$$

gives the so-called Kullback–Leibler divergence.

Proof. (a) The activation function has to be the softmax function. In the end, we require a normalised probability distribution - that is, the predicted probabilities have to sum up to 1.

Additionally, we would probably require differentiability in backpropagation. The standard function that satisfies all these properties is the softmax function.

(b) Over infinitely many large snapshots, since the q_is are constant, we would only average the t_is, as the expression is linear. Thus, we would have

$$\begin{aligned} H &= -\frac{1}{N}\sum_{i=1}^{N}(\langle t_i\rangle \log_2(q_i) + (1 - \langle t_i\rangle)\log_2(1-q_i)) \\ &= -\frac{1}{N}\sum_{i=1}^{N}(p_i \log_2(q_i) + (1-p_i)\log_2(1-q_i)) \end{aligned}$$

(c) By the central limit theorem, we expect the expectation value of t_i to approach p_i, in the sense that the distribution of $\langle t_i \rangle$ should converge in distribution to a delta distribution with support only at p_i. Thus, since H is linear in the t_i, as the number of samples increases, we expect $H_S \approx H$ (again, where the convergence is in distribution).

(d) We compute the partial derivative of this with respect to each q_i:

$$\begin{aligned} \frac{\partial H}{\partial q_i} &= -\frac{1}{N \ln 2}\left[\frac{p_i}{q_i} - \frac{1-p_i}{1-q_i}\right] = 0 \\ \frac{p_i}{q_i} &= \frac{1-p_i}{1-q_i} \\ p_i(1-q_i) &= q_i(1-p_i) \\ p_i &= q_i \end{aligned}$$

To show that it is indeed a minimum, we must compute the second partial derivative

$$\frac{\partial^2 H}{\partial q_i^2} = \frac{1}{N \ln 2}\left[\frac{p_i}{q_i^2} + \frac{1-p_i}{(1-q_i)^2}\right] > 0.$$

(e) We begin by quoting the definition of the Kullback-Leibler divergence on wikipedia as referred to in the lecture notes

$$D_{\mathrm{KL}}(P||Q) = \sum_{x \in \chi} P(x) \log_2 \frac{P(x)}{Q(x)}$$

where $P(x)$ is the real probability distribution. Since there are only two possible events, the sample space is $\chi = \{0, 1\}$ and we have $P(1) = 1 - P(0)$. Defining $P(1) = p$, this means that the Kullback-Leibler divergence can be written as

$$D_{\mathrm{KL}}(P||Q) = p \log_q \frac{p}{q} + (1-p) \log_2 \frac{1-p}{1-q}.$$

We compute the difference

$$\begin{aligned} H - H_{\min} = &- \frac{1}{N} \sum_{i=1}^{N} (p_i \log_2(q_i) + (1-p_i) \log_2(1-q_i)) \\ &+ \frac{1}{N} \sum_{i=1}^{N} (p_i \log_2(p_i) + (1-p_i) \log_2(1-p_i)) \\ = &\frac{1}{N} \sum_{i=1}^{N} \left[p_i \log_2 \left(\frac{p_i}{q_i} \right) + (1-p_i) \log_2 \left(\frac{1-p_i}{1-q_i} \right) \right] \end{aligned}$$

which is the same expression. □

Problem 2 (Cross Entropy)**.** In the previous exercise, we considered probabilities of binary events (light or no light).

Let us now consider M different possible events indexed by $\mu = 1, \ldots, M$. Let us assume that these outcomes occur with the (true) probabilities $p_\mu(\vec{x})$. We would like to construct a network predicting these probabilities.

(a) Let $q_\mu(\vec{x})$ be the probabilities predicted by the network. Write down the cross entropy loss function.

(b) When minimizing the cross entropy, which constraint has to be taken into account?

(c) Show that the extremum again takes place at the point where the probability distributions p and q coincide. Use the method of Lagrange multipliers to take care of the constraint in (b).

Proof. (a) As given in the lecture notes, the cross entropy is given by

$$\mathcal{E} = -\sum_{\mu=1}^{M} p_\mu(\vec{x}) \ln q_\mu(\vec{x}).$$

(b) The constraint is the normalisation of the probability distribution

$$\sum_{\mu=1}^{M} q_\mu(\vec{x}) = 1.$$

(c) The parameters are the q_μ. Thus, we introduce a Lagrange multiplier λ and optimise with respect to $\{q_\mu, \lambda\}$ the auxilliary function

$$g(\{q_\mu\}, \lambda) = -\sum_{\mu=1}^{M} p_\mu(\vec{x}) \ln q_\mu(\vec{x}) + \lambda \left[\sum_{\mu=1}^{M} q_\mu(\vec{x}) - 1\right].$$

As usual, the λ derivative yields the constraint equation. The other derivatives yield

$$\frac{\partial}{\partial q_\mu} g(\{q_\mu\}, \lambda) = -\frac{p_\mu(\vec{x})}{q_\mu(\vec{x})} + \lambda = 0.$$

This has the easy solution

$$q_\mu(\vec{x}) = \frac{1}{\lambda} p_\mu(\vec{x}).$$

Now, since we have the normalisation condition

$$\sum_{\mu=1}^{M} p_\mu(\vec{x}) = 1,$$

the only possibility for this to be compatible with the constraint equation for q_μ is that $\lambda = 1$, and thus we have

$$q_\mu(\vec{x}) = p_\mu(\vec{x}), \qquad \mu \in \{1, \ldots, M\}.$$

□